

Technology Stack

Date	15 July 2026
Team ID	SWUID2026234324
Project Name	VideoScriptAI – AI Powered Video Content Generator
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML5, CSS3, JavaScript</i>	<i>Provides a responsive, interactive, and user-friendly interface for collecting user inputs and displaying AI-generated content.</i>

2	Backend / Server-Side	<i>Python (Flask Framework)</i>	<i>Flask is lightweight, easy to develop with, and ideal for handling API requests, routing, and AI integration.</i>
3	Database / Data Storage	<i>No Database (Environment Variables & Runtime Processing)</i>	<i>The application generates content dynamically and does not require persistent database storage. Sensitive API keys are securely stored in the .env file.</i>
4	Cloud / Hosting / Deployment	<i>Local Flask Development Server</i>	<i>Suitable for development, testing, and demonstration of the application before cloud deployment.</i>

5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Enables version control, project collaboration, backup, and source code management throughout development.</i>
6	Third-Party APIs / Other Tools	<i>Groq API (LLaMA 3.3-70B Versatile), Flask-CORS, python-dotenv</i>	<i>Groq API powers AI content generation, Flask-CORS enables frontend-backend communication, and python-dotenv securely manages API credentials.</i>